

Encounter Databases

A Plain-Language Guide

The everyday business of billing and claims turns out to be one of the richest data sources in drug safety research — if you know how to read it.

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What Is an Encounter Database?

Encounter databases hold electronic records of healthcare encounters for large, defined populations — patient characteristics, prescription fills, and medical services — captured as part of the routine administration or reimbursement of care.

They've become a cornerstone of drug-safety research. An increasing number are available worldwide, and despite wide differences between them, they share a set of defining features worth understanding as a group.

A Population You Can Count

The one essential feature of every encounter database used for research: it is **population based**. Healthcare services are recorded for a defined group of people, regardless of which provider treats them or where they're treated.

That precise definition is what avoids selection bias — the kind that creeps in when, say, a hospital-based study only sees patients who happened to show up at that one hospital. It's also what makes complete, unbiased capture of care possible.

The population doesn't need to represent an entire country to qualify — it just needs to be accurately and completely defined.

FOUNDATIONS, 3 OF 5

Four Domains of Data

Encounter databases useful for drug-safety research typically draw on four core data domains, each capturing a different slice of a person's healthcare life.

Membership

Who's covered and when: patient ID, sex, age, zip code, enrollment and disenrollment dates, and often race/ethnicity or benefit category.

Outpatient Medical

Office and clinic visits: date, location, procedure and diagnosis codes, and the treating provider's identity and specialty.

Inpatient Medical

Hospital stays: primary and secondary diagnoses, admission and discharge dates — usually a summary of the whole stay, not day-by-day detail.

Pharmacy

Outpatient drug dispensings: drug identity, dose and strength, date and quantity dispensed, and days of supply.

Typically not recorded: the reason a drug was prescribed, medication given during a hospital stay, or over-the-counter purchases.

One Database, or Several Linked Together

Some encounter databases keep all four data domains in separate files inside one **integrated** system — most US private and government databases work this way.

Others exist as a **federated** network: independent databases, often run by different organizations, that function together as one virtual database. The Nordic healthcare databases are a leading example.

Either way, the database needs reliable linkage — tying one person's records together across files and over time. And because this data is fundamentally administrative, validating it against an outside source is essential before trusting what it implies about someone's medical history.

Encounter Data vs. Electronic Health Records

This is a different kind of data from electronic health record (EHR) databases. EHRs are built primarily to support patient care; encounter databases arise as a byproduct of administering or paying for healthcare.

That distinction shapes accuracy. Claims tied directly to a payment — the drug dispensed, the procedure billed — are routinely audited and highly accurate. Details that don't affect payment, like the specific reason for a visit, are recorded with far less rigor.

This guide focuses on what encounter databases have in common and how they differ, illustrated with real examples — not on cataloguing every individual database in encyclopedic detail.

What Makes One Database Different From Another?

Encounter databases share a basic set of defining characteristics, but they differ enormously in the attributes that determine whether one fits a given research question.

Some of the biggest differences: how many people are in it and how stable their coverage is, which services get captured and how completely, whether it can be linked to outside data, and how hard it is to actually get access.

KEY ATTRIBUTES, 2 OF 9

Size Isn't Just Size

A large study population is usually necessary for adequate statistical power — especially when exposures or outcomes are rare, effect sizes are small, or subgroup effects matter.

But some of the very design choices meant to strengthen a study also shrink it. A new-user active-comparator design keeps only a fraction of a drug's total users. Restriction, a common way to reduce confounding, can substantially cut cohort size. Instrumental variable methods are statistically inefficient compared with standard regression.

So the number that matters isn't the size of the overall database — it's the size of the actual study cohort left after every one of these choices is applied. The same principle cuts within a single database, too: pharmacy coverage, dental benefits, capitated arrangements, chart-retrieval rights, and linkage permission may each reach only a subset of members, so suitability should be judged for that subset, not the database's nominal size.

KEY ATTRIBUTES, 3 OF 9

A Whole Nation, or a Shifting Slice of It

Universal Coverage

Taiwan, South Korea, Canadian provinces, and northern Europe include their entire population with no eligibility restrictions. Membership is stable for life, so the database tracks the country as a whole.

Fragmented Coverage

The US, above all, produces databases that are far more heterogeneous — each one covering a distinctly different, shifting subset of the population.

KEY ATTRIBUTES, 4 OF 9

Churning, and the Coverage-Gap Problem

People move between databases as their job, income, or residence changes — producing short average enrollment periods, or **dwell time**. Short dwell time makes it harder to tell new drug users from existing ones, since some lag at the start of a person's record is unavoidable.

The time period a database covers matters too. The US Medicaid Analytic eXtract (MAX), for example, carries roughly a three-year lag before data become available — workable for long-term utilization trends, but not for studying a medicine newly on the market.

KEY ATTRIBUTES, 5 OF 9

The Pharmacy Baseline

Medication data is a prerequisite for virtually every encounter-database study — and it's generally limited to drugs dispensed by community pharmacies.

Drugs given during a hospital stay, in an emergency room, or in an outpatient office are typically not captured this way, though they're sometimes reachable through drug-specific procedure codes. Over-the-counter use generally isn't recorded at all, unless it happens to be prescribed and covered by insurance.

KEY ATTRIBUTES, 6 OF 9

When a Drug Purchase Goes Uncounted

Even pharmacy dispensing has blind spots — whenever a drug isn't billed through the insurance system, it can vanish from the record.

Cash Beats Copay

If a drug's cash price is lower than the copay, patients often pay out of pocket and no claim is ever filed — though studies find limited real-world impact from low-cost generic programs.

Excluded From Coverage

A drug can be excluded from reimbursement entirely — benzodiazepines were excluded from Medicare Part D until 2013, invisible to any Part D-based study before then.

Formularies and Prior Authorization

Formulary rules, step therapy, and prior-authorization requirements restrict what's covered and captured — sometimes database-wide, sometimes plan by plan.

KEY ATTRIBUTES, 7 OF 9

How Much Detail Hospital and Outpatient Records Hold

Hospital services are usually recorded as discharge-level data — a summary of a whole stay or emergency visit, not each individual service. Databases differ in diagnosis fields and whether procedure codes exist at all.

Nordic databases capture hospital outpatient and specialist care, but generally lack detailed diagnosis and laboratory information from general-practice visits. GP contacts can often be identified, but not the reason for the visit, so medication use may sometimes serve as a proxy for outpatient diagnoses.

Dental, vision, and long-term care capture vary by database and by a person's own plan; some benefit packages "carve out" services like mental health entirely. Race, ethnicity, and provider-identity data exist in some databases and are simply absent from others.

KEY ATTRIBUTES, 8 OF 9

Linking Out: To Validate, or to Supplement

Linkage to outside data serves two purposes: validating encounter-based information against a gold standard like the medical record, or supplying variables the database lacks — lab results, lifestyle factors, disease severity.

Linking itself comes in two flavors. Deterministic linkage requires an exact match on a unique identifier or an exact combination of several identifying variables — reliable, but it misses imperfect matches. Probabilistic linkage tolerates imperfect matches, catching more true pairs but risking a few false ones. The Nordic prescription-registry networks lean on deterministic linkage; the Dutch PHARMO system uses probabilistic methods instead.

KEY ATTRIBUTES, 9 OF 9

Getting Your Hands on the Data

Access rules, cost, and timelines vary enormously and often drive database choice as much as anything else. Some databases hand over an anonymized, "off-the-shelf" dataset directly. Others require every analysis to run in-house, through the database's own custodian.

Costs scale with the size and complexity of the request, and in-house analysis adds substantially more. Between application and the start of research, the wait can run from a few weeks to several years — and familiarity with a database is no substitute for checking that it truly fits the question.

WORLD TOUR, 1 OF 13

A World Tour of Encounter Databases

The chapter compares seven types of encounter database — spanning US government and commercial programs, Canadian provinces, northern Europe, and Asia — against the same handful of traits.

Population scope. Relative size. Dwell time. How long data lags before it's available. Ease of access. Ability to validate outcomes. Ability to supplement with outside data. Keep these in mind as the tour moves region by region.

Seven Database Types, Compared

Before the region-by-region detail, here is the shape of the comparison the chapter draws.

Database Type	Example Databases
Government, US	Medicare
Government, US	Medicaid Analytic eXtract (MAX)
Health System, US	Kaiser, Geisinger
Commercial Insurance, US	HealthCore, MarketScan, Optum, PharMetrics
Government, Canada	Saskatchewan, Quebec
Government, Northern Europe	Denmark, Norway, Sweden, Netherlands
Government, Asia	South Korea, Taiwan

The United States: A Fragmented Landscape

In 2016, the US had no single health system — coverage came from a patchwork of employer plans and government programs.



Government coverage splits mainly across Medicaid (62M, 19%), Medicare (53M, 17%), and military health programs (15M, 5%) — and 22% of insured people carried more than one type of coverage at once.

US Private Insurance Databases

Most US healthcare runs through private, largely employer-based insurance. A relatively small number of companies cover most of the privately insured population — Optum, HealthCore, MarketScan, and PharMetrics among them — producing some of the largest research databases available.

But size doesn't mean stability: commercial coverage churns at roughly 30% a year. Only about 50% of members stay continuously covered for a full year, 30% for two years, and just 15% for four or more — and details like formularies and carve-outs vary member to member within the same database.

Integrated Delivery Systems, Like Kaiser

A smaller group of Americans get care through integrated, often not-for-profit delivery systems — Kaiser Permanente is the best-known — that both insure and provide the care themselves, often through their own hospitals.

Because these systems own the patient's entire care, their databases can link the full spectrum of it — encounter data plus complete inpatient and outpatient medical records, with direct access to patients and providers. Many belong to the Health Care System Research Network (HCSRN) and partner with the Sentinel System.

Holding both prescribing and dispensing records also lets these systems study primary nonadherence — whether a written prescription was ever filled at all.

Accessing US Government Data

Medicare and Medicaid data are administered by the Centers for Medicare and Medicaid Services (CMS), which grants research access under a protocol and data-use agreement, reviewed by CMS's own Privacy Board.

The Research Data Assistance Center (ResDAC), based at the University of Minnesota, gives technical support to researchers navigating the process. Fees scale with population size and files requested, and data can be released directly or accessed remotely through the CMS Virtual Research Data Center (VRDC).

US Medicaid

Medicaid covers low-income individuals, administered separately — and differently — by each state. In 2016, average monthly enrollment was 70.9 million: 5.7 million aged, 10.6 million blind or disabled, 28 million children, and 26.7 million other adults, including 11.2 million covered through the ACA's Medicaid expansion.

The Medicaid Analytic eXtract (MAX) organizes claims across five files — person summary, inpatient, long-term care, prescription drug, and other services — built by CMS from state submissions since 1999. That system is now transitioning from MSIS to a richer replacement, T-MSIS.

Coverage doesn't guarantee complete claims. Under capitated managed-care arrangements, individual services can generate less complete encounter data than fee-for-service care — historically, many Medicare and Medicaid studies focused on fee-for-service beneficiaries alone.

US Medicare

Medicare covers nearly everyone 65 and older, plus some people with permanent disabilities, across four parts: A (hospital), B (outpatient medical), C (Medicare Advantage — private-plan bundles), and D (prescription drugs, added in 2006).

57 Million

Average monthly enrollment in 2016 — 48 million aged, 9 million disabled.

35% and 44%

Share of brand-name and generic costs paid by beneficiaries in Part D's 2018 coverage gap, until a \$5,000 out-of-pocket limit.

Coverage doesn't guarantee complete claims. Under capitated managed-care arrangements, individual services can generate less complete encounter data than fee-for-service care — historically, many Medicare and Medicaid studies focused on fee-for-service beneficiaries alone.

WORLD TOUR, 9 OF 13

Canada

Canada's roughly 36 million people all have universal, province-administered coverage. Physician visits, diagnostic tests, and hospitalizations come without payment at the point of care; public drug coverage, though, varies by province.

Each province keeps three linkable databases — beneficiary, medical services, and prescription drugs — but only 7 of 10 provinces have prescription data usable for research. Quebec uniquely caps a study to a random 75% sample of the eligible population, up to 125,000 patients.

The payoff for smaller databases: real stability. A study of benzodiazepines and Alzheimer's disease had a full 10-year follow-up available.

The Nordic Countries

Denmark, Iceland, Norway, Sweden, and Finland — a combined population over 25 million — offer tax-supported universal coverage. Pharmacies have submitted dispensing data electronically since somewhere between 1994 and 2006, depending on the country, all linked by a unique civil registration code.

Hospital discharge registries reach back further in some countries than others — Finland to 1969, Denmark to 1977, Sweden to 1987, Norway to 2008. Danish prescription data is especially restrictive: it cannot leave the country's own "data haven," so multinational studies pool results through Statistics Denmark instead of sharing raw records.

More of Europe

PHARMO (Netherlands)

A pharmacy-based federated network limited to specific regions, able to link to outcome and confounder data much like the Nordic model.

Tayside MEMO (Scotland)

A similar regional pharmacy-based network, alongside integrated encounter databases in parts of France and Italy.

SNIIRAM (France)

France's national claims database captures over 66 million people — about 98% of the population — though access to it is complex.

WORLD TOUR, 12 OF 13

South Korea and Taiwan

Both run single-payer, government-administered systems with established national research databases. South Korea, Taiwan, and Australia all code drugs the same way (ATC), and all but Taiwan share the same diagnosis coding (ICD-10) — consistency that lets the Asian Pharmacoepidemiology Network (AsPEN) run studies across countries at once.

Taiwan's NHIRD

Established 1995; covers over 23 million people, more than 99% of the population; updated twice a year.

South Korea's HIRA

Roughly 50 million people; public for research since 2009; access limited to a rolling five-year window. Diagnostic validity runs higher for more severe conditions and for inpatient versus outpatient diagnoses.

Australia, and How Accessible Is Accessible?

Australia's commonwealth government maintains a 10% sample of subsidized medicine and medical-service claims, linked longitudinally, plus separate data covering military veterans, their dependents, and spouses.

A study of 54 Asia-Pacific databases found very few allowed access to raw data at all. Australia, Taiwan, and Japan rated high accessibility; South Korea, medium; Thailand, China, Malaysia, and Singapore, low — a reminder that consistent coding doesn't guarantee consistent access.

STRENGTHS, 1 OF 7

Six Strengths of Encounter Databases

Encounter databases have real strengths compared with other data sources for drug-safety research — which is exactly why they show up so often in the published literature.

Speed and scale. Support for every study design. Linkage to outside data. Broad representativeness. Accuracy of what was actually paid for. And data collection that's blind to the hypothesis being tested.

STRENGTHS, 2 OF 7

Strength One: Fast, Large Cohorts

Automated healthcare databases assemble extremely large cohorts rapidly and cheaply. Several databases in this guide cumulatively cover more than 100 million people. One study shows just how much larger that makes them than an EHR database, even the biggest one:

2.2M

MarketScan

1.5M

Canadian provinces, combined

0.6M

UK GPRD, the largest EHR database

GPRD's cohort excluded patients 65 and older, who made up roughly 35% of the study population in the other databases. Even so, MarketScan's cohort ran more than 3.5 times larger.

STRENGTHS, 3 OF 7

Strength Two: Supports Every Study Design

Because encounter databases are population based and capture every covered encounter regardless of provider, they can support the full range of epidemiologic study designs — cohort, nested case-control, and self-controlled designs alike.

That's a real limitation for non-population-based data, like an EHR database built from a single institution or health system — it simply can't support the same range of designs.

STRENGTHS, 4 OF 7

Strength Three: Linkage to Outside Data

Many large encounter databases can be linked, systematically or on an ad hoc basis, to medical records, disease registries, lab results, and patient or provider surveys.

In ideal circumstances, this combines the best of both worlds: the size and population-based reach of encounter data, with the clinical detail and higher quality of the linked source. But linkage ability and quality vary substantially between databases and have to be weighed for each research question.

STRENGTHS, 5 OF 7

Strength Four: Broad Representativeness

Many large encounter databases are broadly representative of a nation, a region, or a particular health system — well suited to health-services and health-policy research.

Many also include very stable populations, useful for assessing long-term safety and long-term trends in treatment practice. And databases from countries with universal coverage are — by definition — free from selection bias, since everyone is included.

STRENGTHS, 6 OF 7

Strength Five: Accuracy of What Was Paid For

Data elements tied directly to a payment amount — the procedure billed, the drug and quantity dispensed — are subject to auditing and highly accurate.

But accuracy of the procedure only confirms it happened, not why. One validation study used surgical codes in Medicaid data to identify hip-fracture cases; the medical record confirmed every procedure had truly been performed — but some had been done to correct a different orthopedic condition entirely.

Pharmacy dispensing data has a similar edge over prescriptions recorded in EHRs: a dispensing is one step closer to actual ingestion than a prescription is, so it carries less exposure misclassification.

STRENGTHS, 7 OF 7

Strength Six: Blind to the Hypothesis

Accuracy for data not tied to payment is less well established, and depends heavily on the rigor of each database's own quality-assurance processes.

But there's a compensating strength: data capture in encounter databases is automated and completely independent of the study question or hypothesis being tested. That sharply reduces the risk of recall or assessment bias — the kind that creeps in when data collection knows what a study is looking for.

LIMITATIONS, 1 OF 6

Five Limitations to Keep in Mind

Encounter databases are built and maintained primarily for payment and administration, not research — and that shows up as real limitations once the data is put to scientific use.

Uncertain diagnostic validity. Missing clinical detail. The real cost of fixing either through linkage. Exposure data that can miss in both directions. And, in the US specifically, fragmentation that undercuts representativeness.

LIMITATIONS, 2 OF 6

Limitation One: Uncertain Diagnostic Validity

One of the biggest concerns in using encounter data for research is the uncertain validity of diagnostic codes — a concern that applies broadly, but is amplified in outpatient settings, where a diagnosis usually isn't tied to a specific level of payment.

That makes it critical to validate diagnostic data — for both outcomes and important confounders — against an external gold standard, like the medical record or a disease registry. Even those gold standards may not be perfectly correct compared with a randomized trial's own diagnoses.

LIMITATIONS, 3 OF 6

Limitation Two: Missing Clinical Detail

Encounter data lack markers of disease severity (blood pressure, ejection fraction) and lifestyle factors (smoking, alcohol use, body mass index, physical activity). Diagnostic codes for things like obesity sometimes exist, but with very low sensitivity — one validation study found claims-based codes caught only 18% of Medicare patients who actually had obesity.

Because this clinical detail is so often critical for confounding adjustment, methods built to minimize unmeasured confounding — self-controlled designs, active-comparator new-user designs, instrumental variable analysis, propensity score calibration — become especially important for encounter-based research.

LIMITATIONS, 4 OF 6

Limitation Three: Linkage Has Its Own Costs

Linking to outside data like EHRs, disease registries, or lab results can overcome many of these gaps — but the linkage itself is typically time-consuming and costly, and often available for only a subset of the database's population.

The resulting enriched dataset also tends to remain less comprehensive than a true population-based EHR database, and validation of the linkage is often limited to small samples with poor response or retrieval rates.

LIMITATIONS, 5 OF 6

Limitation Four: Exposure Data Can Miss Both Ways

Claims Miss the Unbilled

Drugs excluded from reimbursement, bought over the counter, or paid for in cash because it's cheaper than the copay can all vanish from claims — making an exposed patient look unexposed.

EHRs Miss the Undispensed

EHR prescription records capture nonreimbursable and low-cost generic drugs better, since they log everything written — but not everything written gets filled, which can make an unexposed patient look exposed.

LIMITATIONS, 6 OF 6

Limitation Five: US Fragmentation

Because the US healthcare system is so fragmented, many of its largest encounter databases fall short of representing the general population — and feature significant turnover and short dwell times.

US private insurance databases and the Medicaid Analytic eXtract (MAX) are both examples: valuable at massive scale, but built from a population that keeps shifting underneath the data.

PUTTING TO WORK, 1 OF 3

Putting a Database to Work

Encounter databases have supported thousands of published studies — in drug safety, comparative effectiveness, drug utilization, health services research, methods, and pharmacoeconomics. A handful of everyday steps show up across virtually all of them.

Two are close to universal: linking records together across files and over time, and turning raw codes into the study variables that actually answer a research question.

PUTTING TO WORK, 2 OF 3

Turning Codes Into Study Variables

Virtually every study requires linking records between data files and over time, so that an individual's entire set of encounters across the study period can be analyzed together.

The other near-universal step is aggregating drug, diagnosis, and procedure codes into meaningful variables. Exposures, outcomes, confounders, and inclusion criteria are all defined through code lists, usually specific to both the study and the database — ideally reusing algorithms already validated elsewhere.

Coding systems change over time, too. A validated algorithm under one code set may not remain valid after a transition, and studies spanning a coding change may need multiple code lists and renewed validation.

PUTTING TO WORK, 3 OF 3

Building the Study Population and Its Timeline

Together with demographic information, these study-specific variables create the study population — often (new) users of a specific drug or drug class among people who meet the study's inclusion and exclusion criteria.

From there, exposure, outcomes, and confounders are measured over time relative to the study's index date, letting researchers calculate person-time at risk, incidence rates, and measures of association. When the database still lacks something needed, researchers reach outside it — to linked EHRs, direct surveys, paper charts, or disease and immunization registries.

Choosing the Right Database

This is genuinely new ground for this edition of the textbook: a dedicated discussion of how to evaluate whether a specific encounter database — or which of several candidates — actually fits a given research question.

Five questions recur across real published examples: Does it capture the target population? Is the usable cohort big enough? Can the outcome be validated? Does it have the specific service needed? And can it be supplemented where it falls short?

CHOOSING A DATABASE, 2 OF 6

Does It Capture Your Target Population?

Stroup and colleagues wanted to compare clozapine against a standard antipsychotic in adults with treatment-resistant schizophrenia. National US Medicaid data wasn't the obvious choice — Medicaid's population is often selective and transient.

But Medicaid covers about two-thirds of all US adults with schizophrenia, and because most qualify through disability rather than economic need, they're typically enrolled stably, without gaps. Their 1:1 propensity-matched cohort study found clozapine lowered the risk of psychiatric hospital admission (hazard ratio 0.78) but raised the risk of diabetes (hazard ratio 1.63).

CHOOSING A DATABASE, 3 OF 6

Is the Usable Cohort Big Enough?

Shin and colleagues studied the risk of cardiovascular events in children and adolescents with ADHD taking methylphenidate. Because the outcome is rare, they turned to South Korea's HIRA database, with more than 50 million people.

From that population, 144,258 children had an ADHD diagnosis; 114,657 were new users of methylphenidate, and just 1,224 had an incident cardiovascular event. With an outcome this rare, a self-controlled case series design was used instead — needing far fewer patients than most designs for similar power.

CHOOSING A DATABASE, 4 OF 6

Can You Validate the Outcome?

Lo Re and colleagues ran parallel safety studies of saxagliptin across four databases — two EHR databases and two encounter databases, Medicare and HealthCore. One requirement for the encounter databases: the ability to pull inpatient records to confirm each outcome. No increased risk turned up in any of the four.

Some outcomes need a custom algorithm. Using Quebec data, Moride and colleagues built one to detect youth suicide attempts — an injury or intoxication code from an ED visit, followed by a psychiatric consult or diagnosis within two days. It reached 70% sensitivity and 97.6% specificity.

CHOOSING A DATABASE, 5 OF 6

Does It Have the Specific Service You Need?

Not every database captures every kind of care. Gupta and colleagues wanted to examine opioid prescribing among US dentists from 2010 to 2015, using the MarketScan database.

Dental services aren't captured for everyone in that database — so the study population had to be restricted to people with simultaneous enrollment in both a medical and a dental plan.

CHOOSING A DATABASE, 6 OF 6

Can You Supplement What's Missing?

Huybrechts and colleagues compared mortality risk across antipsychotics in nursing-home residents dually eligible for Medicaid and Medicare. Cognitive function and behavioral symptoms of dementia are crucial confounders but poorly captured in encounter data — so the study linked to the Minimum Data Set, a federally mandated nursing-home health assessment. Compared with risperidone, haloperidol carried higher mortality risk and quetiapine carried lower risk.

A Swedish-Danish study did something similar: linking hospital admissions and prescription fills to detailed ECG data let researchers measure not just how often QTc prolongation occurred after suspected poisoning, but how strongly it predicted death within 30 days.

THE FUTURE, 1 OF 3

Where the Field Is Headed

Pharmacoepidemiologic research with encounter databases keeps growing — more databases, in more regions, including places with no data available today.

Three forces are likely to shape what comes next: advancing information technology, tightening privacy regulation, and the fact that this data is a byproduct of healthcare systems that keep changing underneath it.

THE FUTURE, 2 OF 3

Three Forces Shaping the Future

Advancing Technology

Expanding storage and processing will keep opening links to new sources — biomarkers, social media, web searches, wearable biometric data — and make analysis tools more accessible.

Tightening Privacy Rules

The same technology that could build richer data is likely to be counterbalanced by stricter privacy regulation — rules that vary by country and change fast.

Shifting Health Systems

Encounter data is a byproduct of the systems that generate it, built for running healthcare, not research. As those systems change, so does the data.

THE FUTURE, 3 OF 3

The US Shift Toward "Value Health"

US healthcare has historically run on fee-for-service: providers bill for each service, generally justified with a diagnosis, and those paid claims form the core of the encounter databases described here.

But paying more for more care creates an incentive to deliver more of it, driving rapid cost growth. In response, the US is shifting toward a per-patient-per-month "population health" model, with new incentives meant to guard against the opposite risk of under-treatment — sometimes called "value health." Practices and hospitals are consolidating to build the infrastructure this shift requires, and the underlying data is likely to keep changing substantially in the years ahead.

WRAP-UP, 1 OF 2

The Througline

Encounter databases are a byproduct of administering or paying for healthcare, not a purpose-built research tool — and every strength and limitation in this guide traces back to that one fact.

That's exactly what makes them valuable: massive scale, real populations, and data that's accurate for whatever was actually billed. And it's exactly why they demand care — validate what you can, supplement what you can't, and never mistake convenience for completeness.

WRAP-UP, 2 OF 2

Wrapping Up

Encounter-based data is a genuinely important resource for pharmacoepidemiologic research — comprehensive, often high quality, and collected for reasons that have nothing to do with any single study's hypothesis.

But because it's generated for administration, not research, its applicability, completeness, and generalizability always need to be weighed against how convenient it is to use.

And one thing a bigger database never solves on its own: issues of bias and confounding don't shrink just because the sample size grows.